

UNIVERSITY OF ZAMBIA
SCHOOL OF HUMANITIES AND SOCIAL SCIENCES
AUGUST 2016/17 ACADEMIC YEAR FINAL EXAMINATIONS
DEM 1110: INTRODUCTION TO DEMOGRAPHY

SECTION A

ANSWER ALL QUESTIONS IN THIS SECTION

1. Define the following concepts? Give an example for each.
 - a) Continuous and categorical data
 - b) Descriptive statistics
 - c) Statistic and parameter
 - d) Difference between pattern and level
 - e) Primary data collection
 - f) Routine and non-routine data collection
 - g) Passive method of registering vital events
 - h) Crude and specific rates
2. Demonstrate your knowledge of the age structure of the population by answering the following questions:
 - a) In what aging stage is a population with a median age of 23 years?
 - b) How about a population with median age of 18 years?
 - c) And one with median of 37 years?
3. Demonstrate your knowledge of age classifications by indicating the age categories, the following persons belong to:
 - a) A 56 years old man?
 - b) A 10 year old girl?
 - c) A 76 year old woman?
 - d) A 23 year old woman?
 - e) A 17 year old boy?
4. What two major events in Europe influenced Thomas Malthus in writing the book – “An Essay on Population”?
 - a) The rapid increase in the population of the poor in England.
 - b) The opulent life style of the aristocracy in England.
 - c) The outbreak of the Great Plague in Europe.

- d) The deteriorating social conditions of the poor in England.
 - e) The overthrow of the monarchy in France.
5. Which of the following scenarios represent the following: i) Floating population ii) Stagnant population iii) Latent population or vi) None:
- a) Casual workers at Shoprite laid off after going on strike agitating for higher wages.
 - b) ZAMBEEF reduces the number of farm labourers after introducing combine harvesters at their Mpongwe Wheat farm.
 - c) Seasonal farm labourers recruited by Zambia Sugar Company as sugar cane cutters during the harvesting season for sugar cane.
 - d) White commercial farmers in Mkushi Farm Block growing maize for export.
 - e) Wheel barrow drivers and those doing "piece work" in Central Business District of Lusaka.
6. Demonstrate your knowledge of history of demography by answering the questions below:
- a) None of the pioneering contributors to the development of demography was a trained demographer. What were the actual occupations or jobs of the following:
 - i. John Graunt
 - ii. Thomas Malthus
 - iii. Karl Marx
 - iv. Friederich Engels
 - b) Which demographer is reputed to be the intellectual father of ideas of overpopulation and the limits to growth?
 - c) Which demographer pioneered the statistical evaluation of demographic data?
 - d) Which country pioneered the systematic collection of demographical data and established the first statistical office?
 - e) What were the disciplinary backgrounds of the following key and pioneering figures in demography:
 - i. Frank Notestein
 - ii. Raymond Pearl
 - iii. Lowell Reed
 - iv. Alfred Lotka
 - v. Ansley Coale
 - vi. Warren Thompson
 - vii. P. K. Whelpton
 - viii. Louis Dublin
 - f) Apart from the Population Council and the UNFPA, mention any two very important contemporary international organizations involved in the promotion of the scientific research in population?
 - g) What do the following acronyms stand for:
 - i. PAA
 - ii. IUSSP
 - iii. UNZAPOPSA

- h) How many years passed between the establishment of the University of Zambia and introduction of the demography program at the University of Zambia? (Indicate the actual dates as well)
- i) In what year did the demography program at UNZA become a fully-fledged and autonomous department?
- j) Before it became an independent department, the demography programme was part of which department in the School of Humanities and Social Sciences?
7. Which of the statements below best describes i) external demographics ii) cluster marketing iii) targeting iv) segmentation v) cluster marketing vi) none.
- a) Manufacturing and packaging of products or the provision of services that appeal to specific socio demographics identifiable groups within the population.
- b) Picking out particular socio-demographic characteristics of people who might purchase what you have to offer, and then appealing to the consumer tastes and behaviour reflected in those characteristics.
- c) Targeting neighborhoods with a set of shared socio-demographic characteristics.
- d) Analysis of the demographic characteristics of the workers with a view of improving their conditions of service.
- e) Carrying out a market research and then aggressively advertising it on television and radio.
8. On the basis of the data below, compute the relevant rates and briefly interpret the results.
- a) Overall dependency ratio
- b) Child dependency
- c) Aged dependency

Zambia (1980)

Age group	Population
0-14	3,349,672
15-64	3,842,792
65+	190,633

9. Interpret the following demographic characteristics in Zambia.
- a) A net migration rate of +6.7 for Lusaka
- b) A life expectancy of 52 years based on the 2010 Zambian census
- c) An unmet need for family planning of 30 percent in Zambia.
- d) The median age of age 18 years.
- e) A net migration rate of -5.2 for Kaputa

10. The concepts of data and information are sometimes used interchangeably in social sciences. What is the main difference between the two concepts?
 11. As the newly appointed officer responsible for surveys at the Central Statistical Office, outline two (2) conditions where the collection of primary data is appropriate?
 12. The continuous registration of vital events has been seen as a key data source for social and economic planning. Briefly outline three (3) inherent problems of collecting vital events in developing countries?
 13. Several methods are used to visually display data. Give three points to consider when using graphs to visually display data?
 14. Briefly describe what your understanding is of:
 - a) The demographic dividend
 - b) The difference between the rate of population growth and the rate of natural increase
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SECTION B

ANSWER ANY TWO QUESTIONS FROM THIS SECTION

1. As Human Resources Manager at Company X, discuss how the welfare and conditions of service can be influenced by :
 - a) Demographic factors within the company and;
 - b) Demographic factors outside the company
2. Use relevant examples drawn from Zambia to discuss.
 - a) What migration is
 - b) The causes of migration
 - c) The three major types of migration as well as their causes.
 - d) The consequences of migration
3. Answer the following: be brief and concise
 - a) Differentiate between prospective and retrospective method of data collection:
 - b) All methods of data collection are prone to errors; differentiate content and coverage errors:
 - c) Outline three (3) reasons why it is necessary to evaluate data before it is used for social and economic development?
 - d) Give four (4) principle sources of errors in data:

- e) What is the easiest method of identifying errors in Single Year Age and sex data?
 - f) Using the method given in (e) how would you know the data has errors?
 - g) The Myers and Whipples Index are both used to detect age preference, what is the main difference between the methods?
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SECTION C

ANSWER ALL QUESTIONS FROM THIS SECTION

1. Use Table 1 to answer the questions below.
 - a. What does a sex ratio of 105.8 mean for the population aged 35-39 years?
 - b. What is the ratio of the total male population to the female total population?
 - i. What does the result mean?
 - c. What was the proportion of male population above 55 years in Zambia?
 - d. You suspect errors in the age and sex data, what type of error is likely to affect these data in table 1?
 - e. As a Planning Officer at the Ministry of General Education, you have been tasked to plan for Early Childhood Education. What is proportion of children 0-4 years (both sexes) to the total the total population (both sexes)
 - i. What recommendation you give the Minister of Education of the basis of the results in (e)?

Table 1: Population by Five Year Age Group and Sex Ratio, Zambia Census, 2010

Age Group	Population		Sex ratio	Difference
	Male	Female		
0-4	1,102,272	1,112,615	99.1	
5-9	924,395	931,941	99.2	0.1
10-14	840,834	858,208	98.0	-1.2
15-19	692,859	735,025	94.3	-3.7
20-24	504,876	612,600	82.4	-11.8
25-29	464,493	541,751	85.7	3.3
30-34	397,694	403,076	98.7	12.9
35-39	332,961	314,852	105.8	7.1
40-44	232,253	215,331	107.9	2.1
45-49	174,833	180,156	97.0	-10.8
50-54	128,696	141,558	90.9	-6.1
55-59	90,037	94,791	95.0	4.1
60-64	73,419	88,456	83.0	-12.0
65-69	53,695	64,887	82.8	-0.2
70-74	41,929	48,873	85.8	3.0
75+	62,007	64,941	95.5	-
Total	6,117,253	6,409,061	-	

END OF EXAMINATION